Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 03)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 1 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 2 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 2 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

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Answer ☐ t/ha

0 2 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 2 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1 .....

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2 .....

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0 3 . 1 For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases .....

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Whether the numbers of dormice differ between woodlands with different management ..

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Whether mean dissolved oxygen differs between two streams .....

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Whether crop yield increases with fertiliser application (ranked data) .....

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0 3 . 2 Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 3 . 3 A Spearman's rank test on 12 pairs of data gave r_s = -0.72. The critical value at p = 0.05 for n = 12 is 0.59. Explain what can be concluded.

[3 marks]

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0 3 . 4 Explain the purpose of error bars on a graph of mean values.

[2 marks]

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0 4 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

0 4 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer ☐ times

0 4 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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0 4 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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0 4 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

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0 4 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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15

0 5 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 5 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

Do not write outside the box

0 5 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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0 6 . 1 Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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Extra space

0 6 . 2 State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

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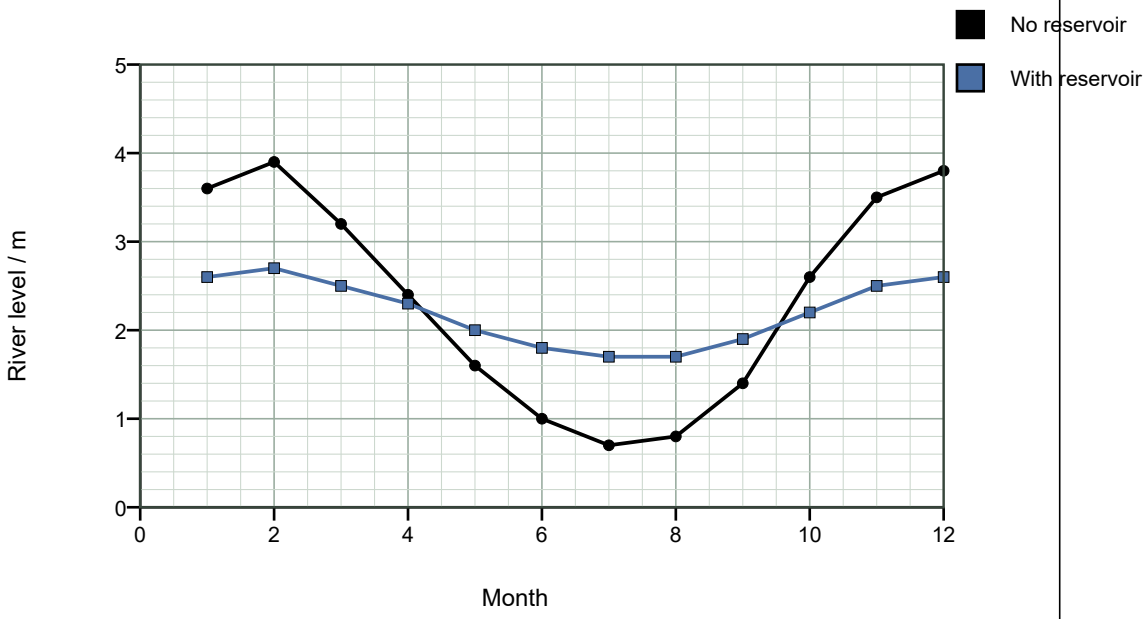
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0 7 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



0 7 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer ..... m

0 7 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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0 7 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

0 7 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

15

0 8 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 8 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 8 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0 9 Figure 1 is a flow diagram of the main stages in a sewage treatment works using activated sludge treatment.

Figure 1

Stage	Process
Pre-treatment	Screens and grit traps
Primary treatment	Primary sedimentation
Secondary treatment	Aeration tank then secondary sedimentation
Tertiary treatment	Phosphate stripping / micro-strainers / UV
Sludge treatment	Anaerobic digestion

0 9 . 1 Explain the purpose of screens and grit traps.

[2 marks]

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0 9 . 2 Explain why primary sedimentation is described as the most important process.

[2 marks]

0 9 . 3 Describe what happens in the aeration tank and explain why the sludge from the secondary sedimentation tank is returned to it.

[3 marks]

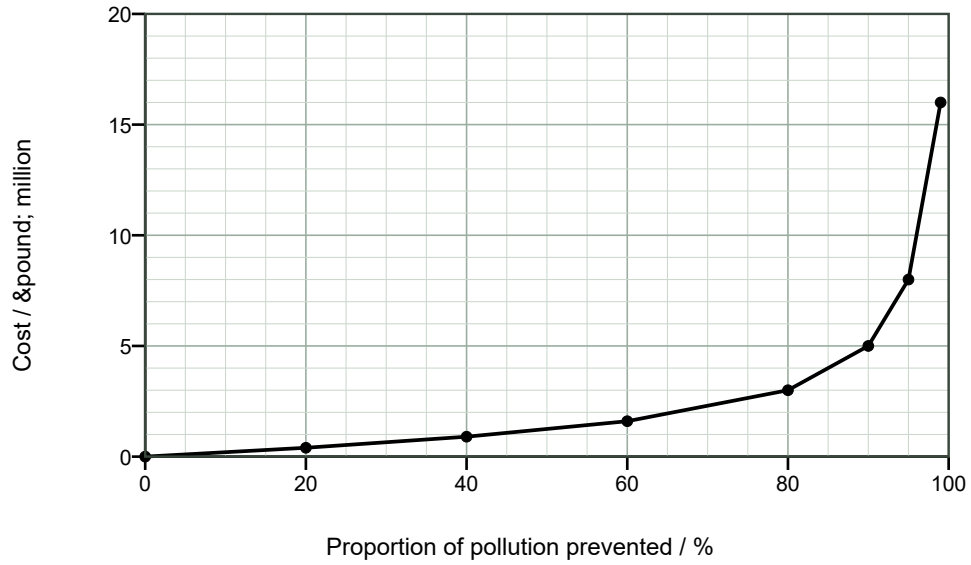
0 9 . 4 Explain why tertiary treatment may be needed before effluent is discharged, and describe **one** tertiary process.

[3 marks]

10

1 0 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



1 0 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer ____ £ million

1 0 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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1 0 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

1 0 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS